UNIVERSITI MALAYA

NFMN

SIM 1005 / SIN 1002

Fundamentals of Spreadsheets / Introduction to Worksheet

Lecture 5: System of Linear Equations

By

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Institute of Mathematical Sciences

Objectives

- To recall some built-in functions for matrix operations.
- To solve system of linear equations by using the inverse method.
- To solve system of linear equations by using Excel Solver.

Original notes by NFM Noor

System of Linear Equations

- A system of linear equations can be represented in matrix form as $Ax = b$ ie.

$$\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$$

- In Excel, this system can be solved using
  1) Inverse method
  2) Excel Solver

Original notes by NFM Noor

Inverse Method

- Given a system of linear equations in matrix form as $Ax = b$.

- If A^{-1} exists then

$$A^{-1}Ax = A^{-1}b$$

- Hence the vector x can be found as

$$x = A^{-1}b$$

Original notes by NFM Noor

Example 1

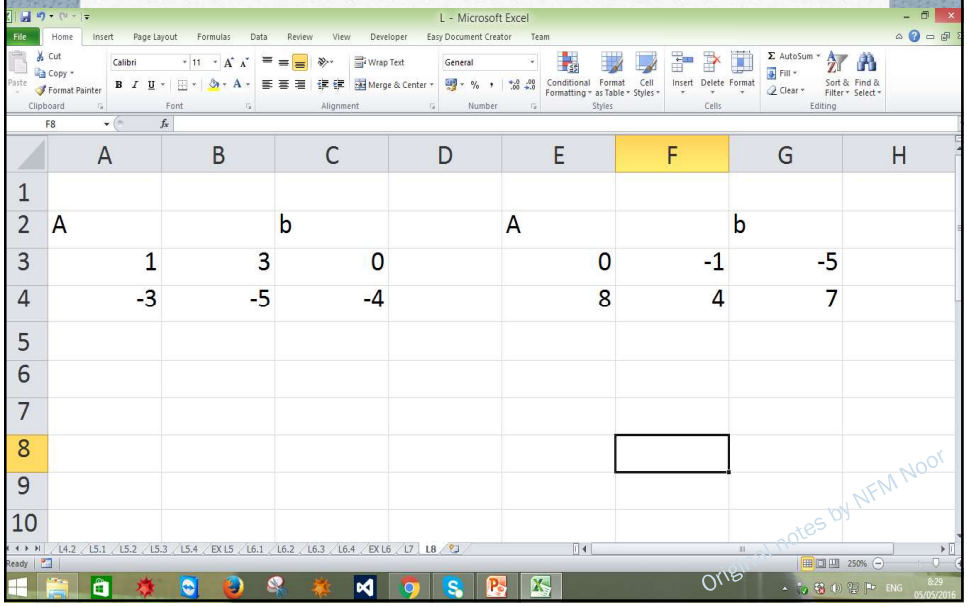
Solve the following systems using inverse method:

1. $\begin{pmatrix} 1 & 3 \\ -3 & -5 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ -4 \end{pmatrix}$

2. $\begin{pmatrix} 0 & -1 \\ 8 & 4 \end{pmatrix}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} -5 \\ 7 \end{pmatrix}$

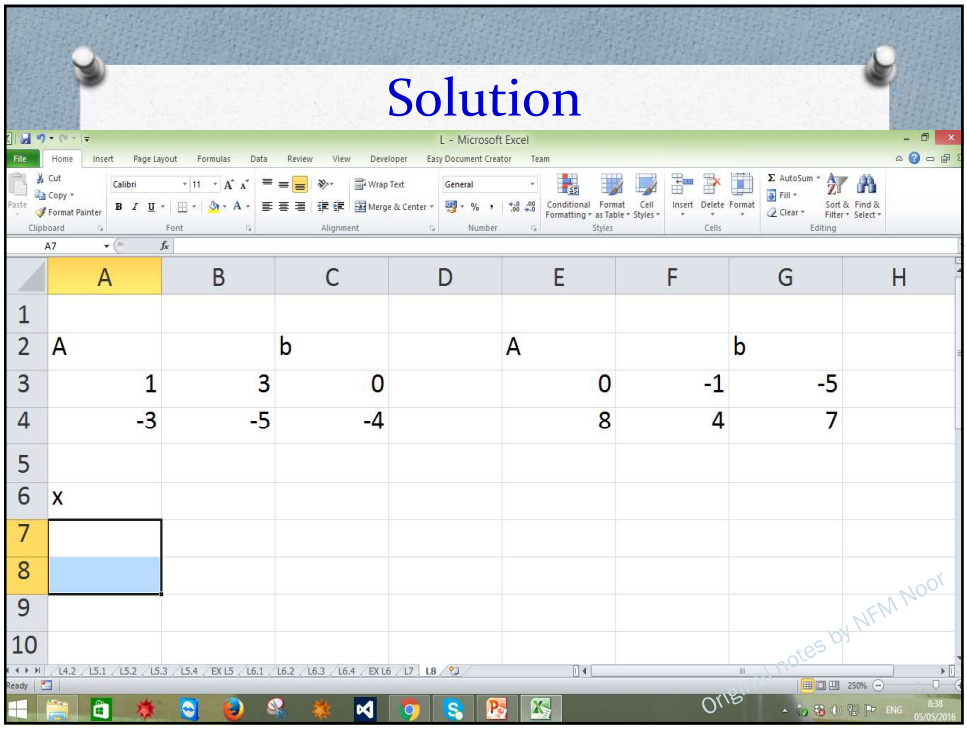
Original notes by NFM Noor

Solution

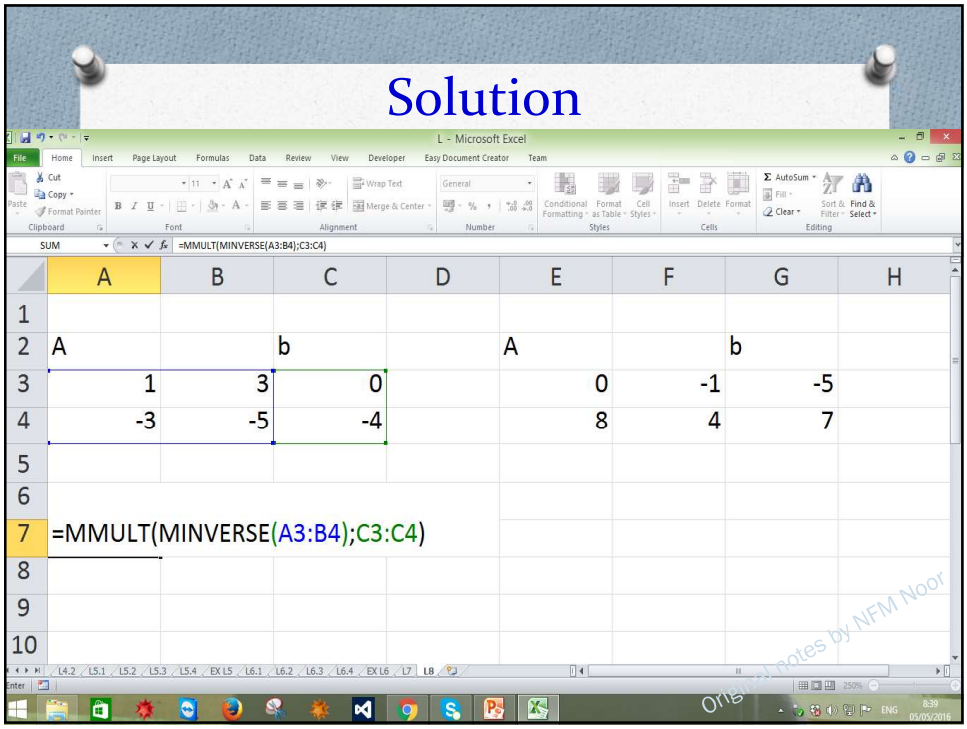


Original notes by NFM Noor

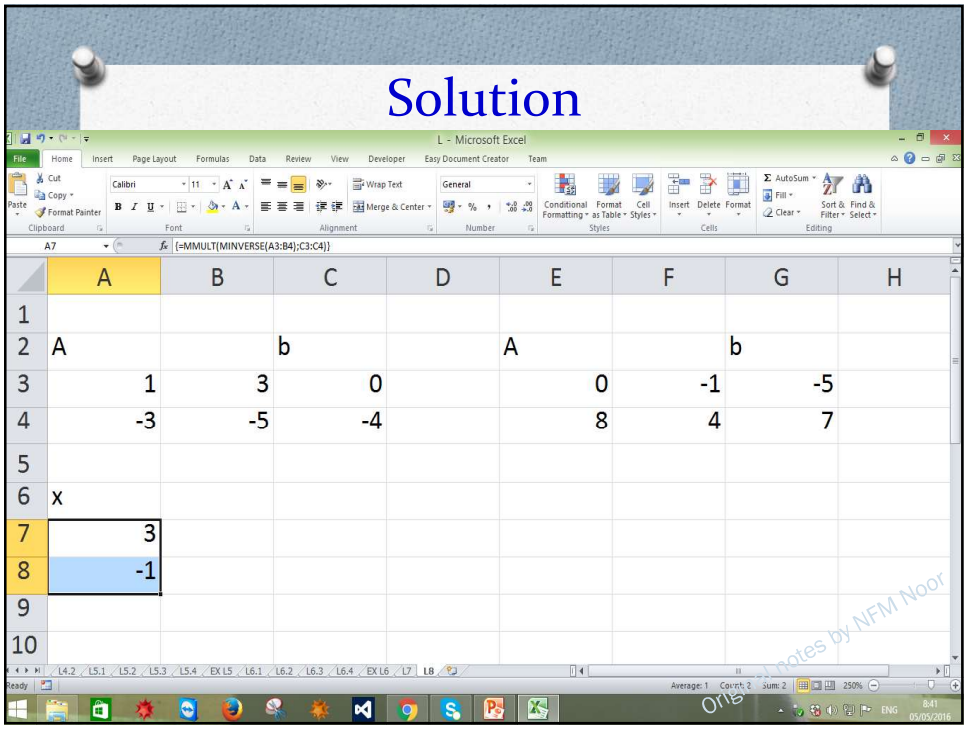
Solution



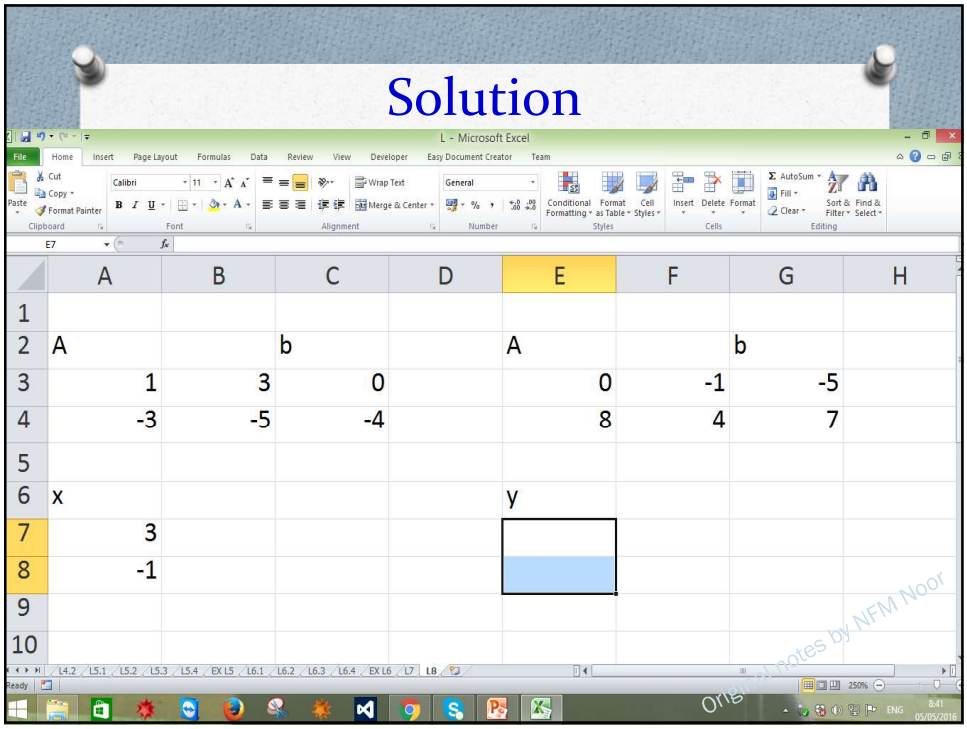
Solution



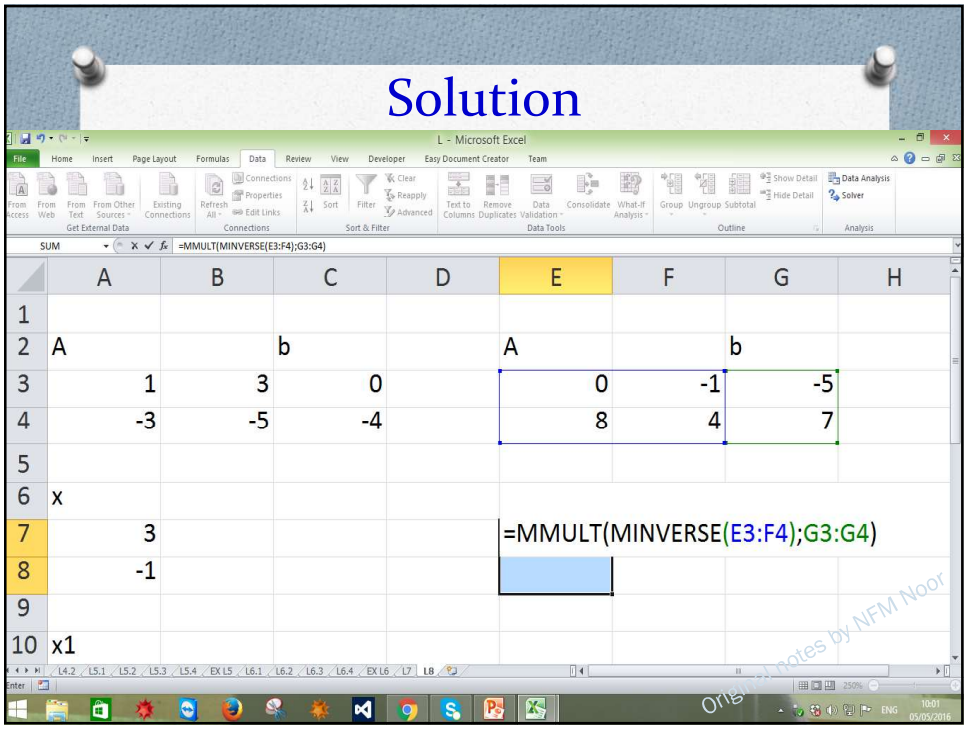
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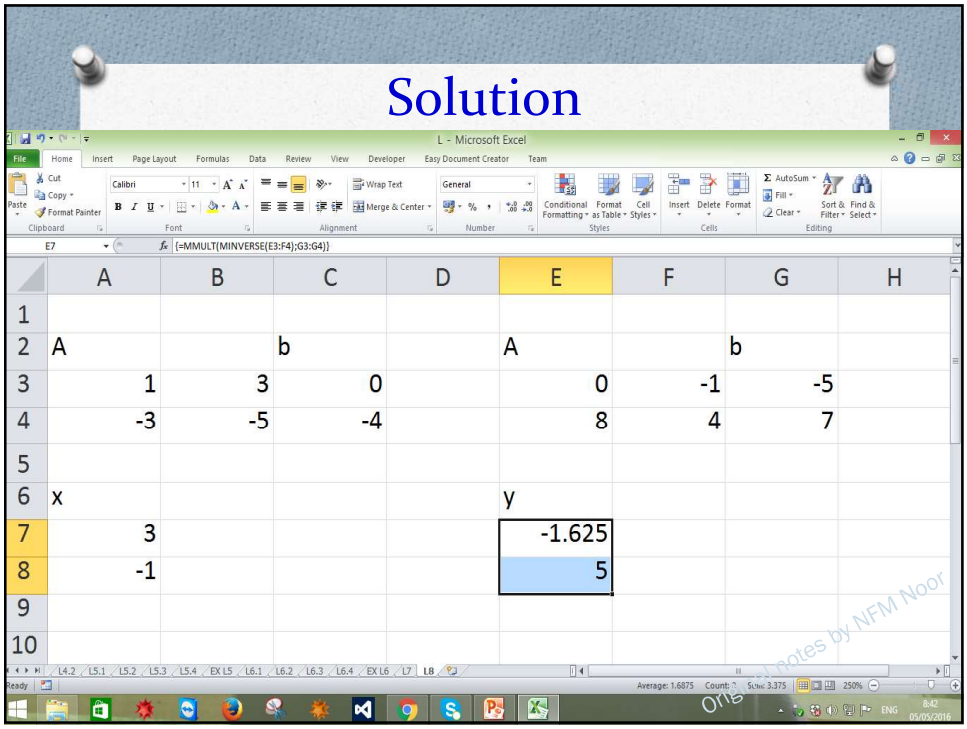
Solution



Solution



Solution



Exercise 1

Solve the following systems using inverse method:

1. $\begin{pmatrix} 4 & -5 & 0 \\ -1 & -2 & 9 \\ 3 & 3 & 8 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 12 \\ 3 \\ -9 \end{pmatrix}$
2. $\begin{pmatrix} -7 & 1 & 2 \\ 10 & 0 & -4 \\ 3 & -5 & -1 \end{pmatrix}\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 7 \end{pmatrix}$

Original notes by NFM Noor

Excel Solver

An Excel optional add-ins analysis tool for solving:

1) System of linear equations
2) Linear programming problem*
3) Operational research problem*

*will not be covered in this course syllabus

Original notes by NFM Noor

Example 2

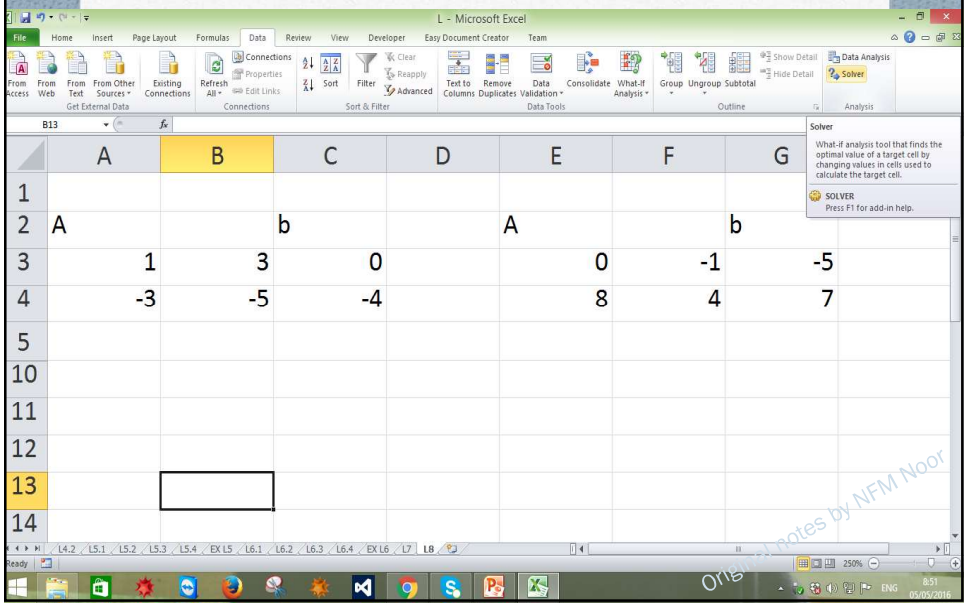
Solve the following systems using Excel Solver:

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2. $\begin{pmatrix} 0 & -1 \\ 8 & 4 \end{pmatrix}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} -5 \\ 7 \end{pmatrix}$

Original notes by NFM Noor

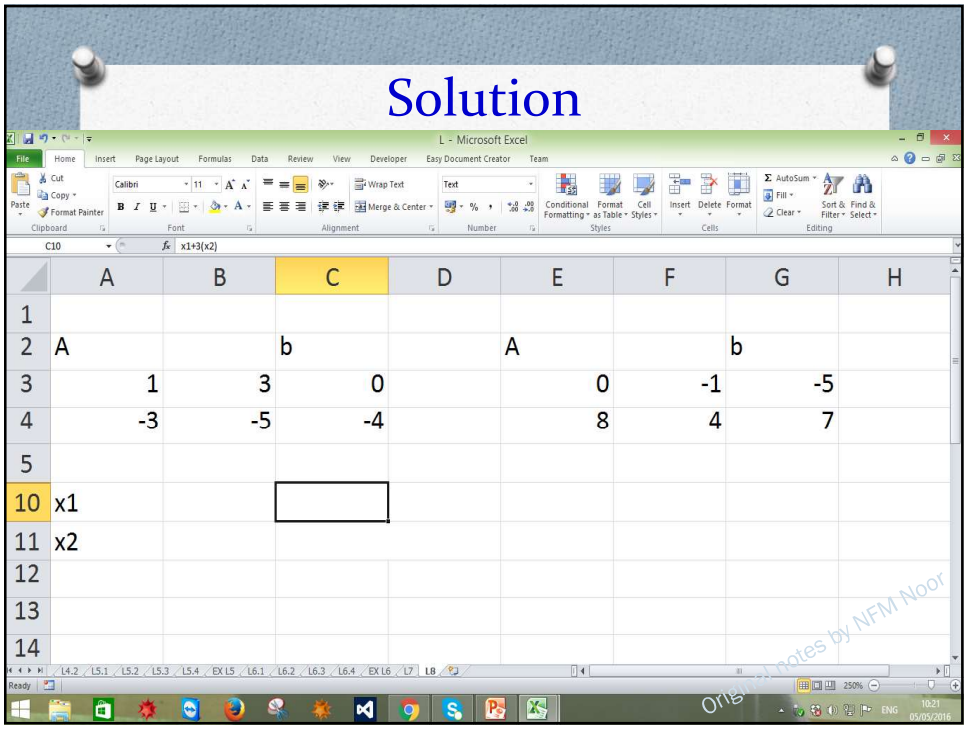
Solution



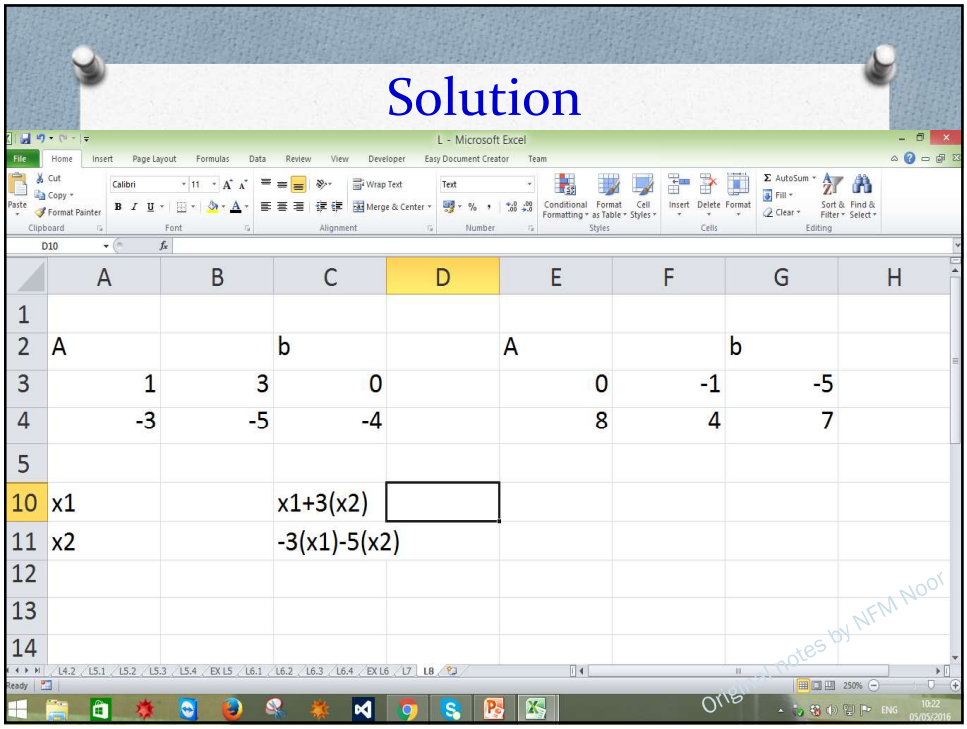
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2	A		b		A		b
3	1	3	0		0	-1	-5
4	-3	-5	-4		8	4	7
5							
10							
11							
12							
13							
14							

Original notes by NFM Noor

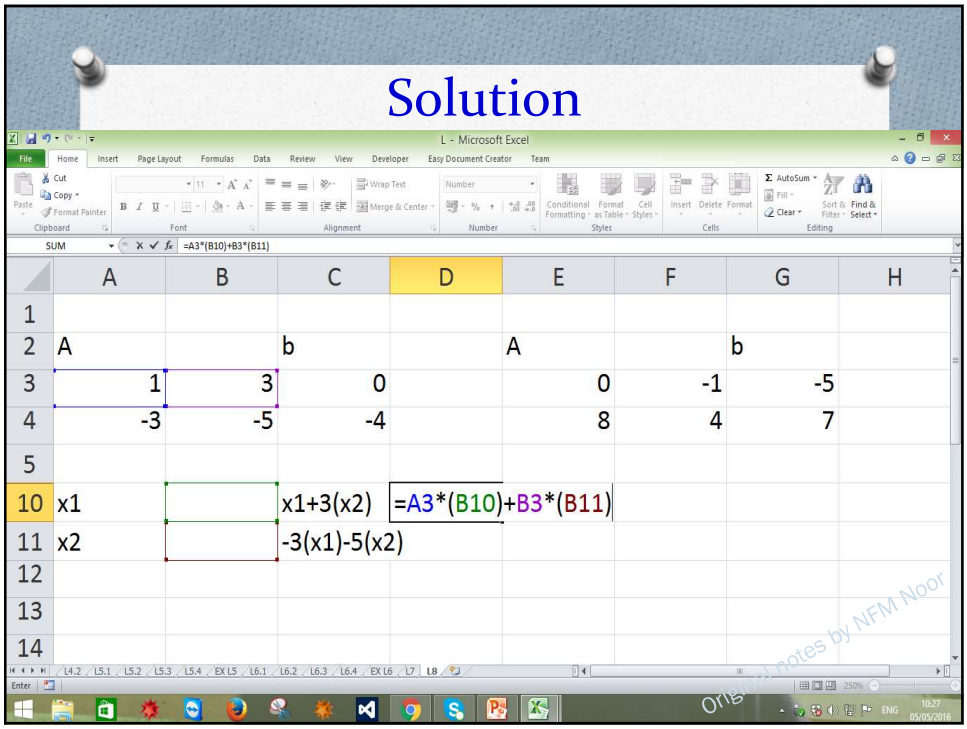
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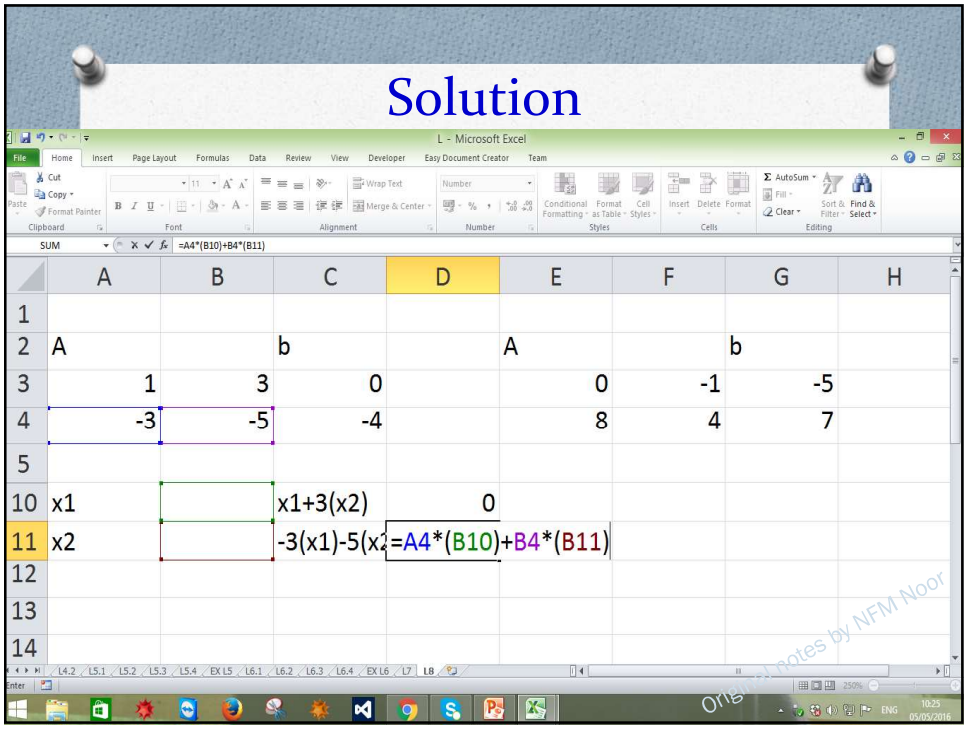
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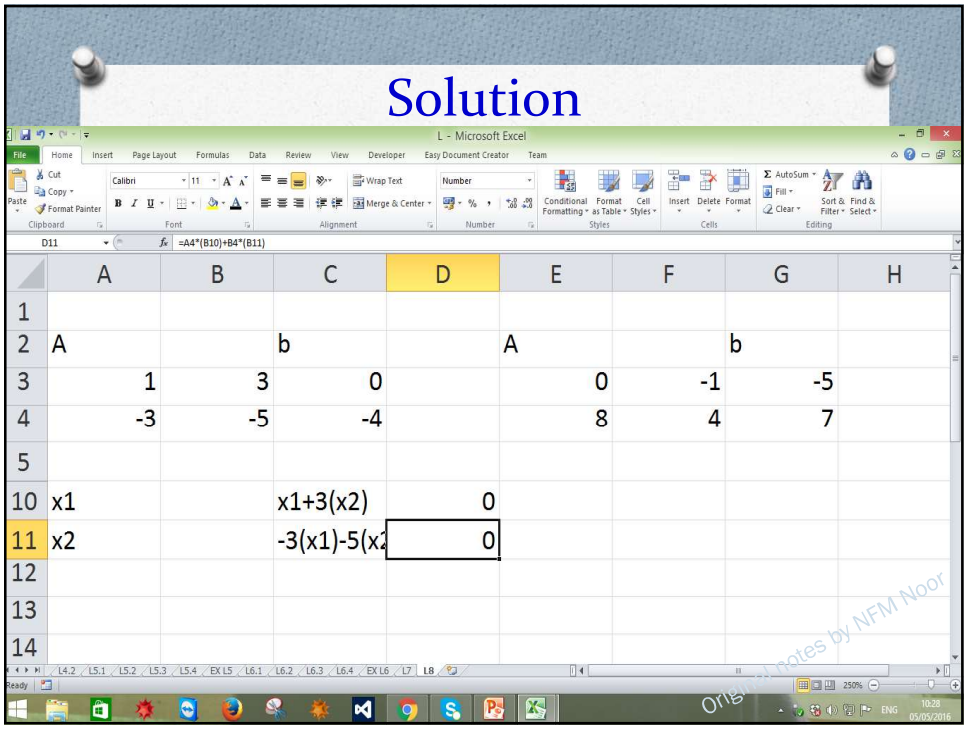
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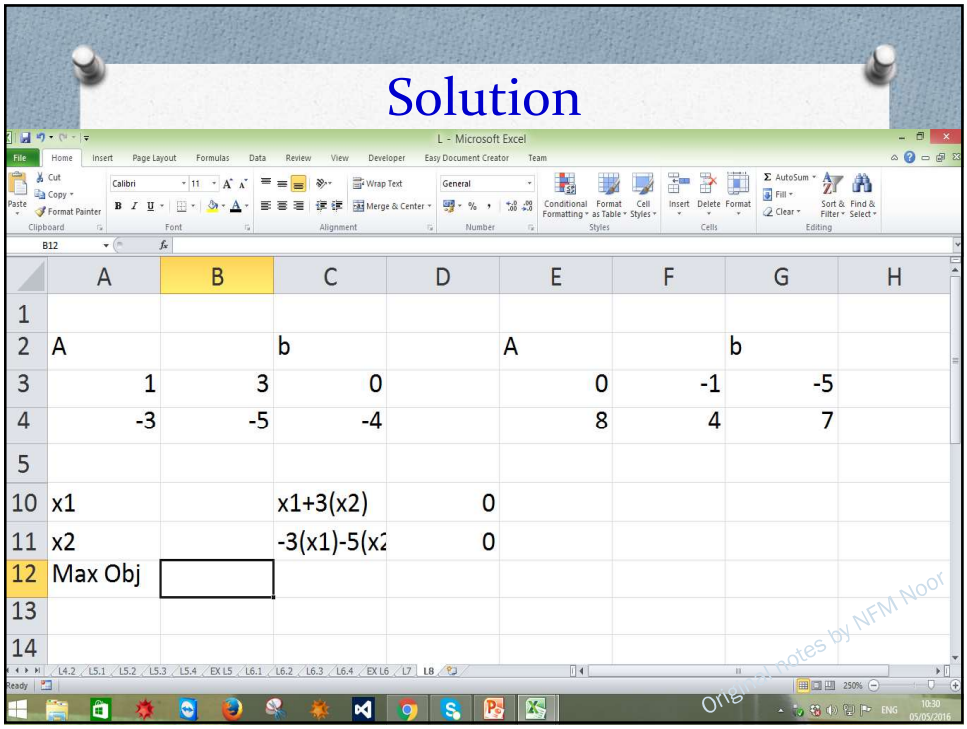
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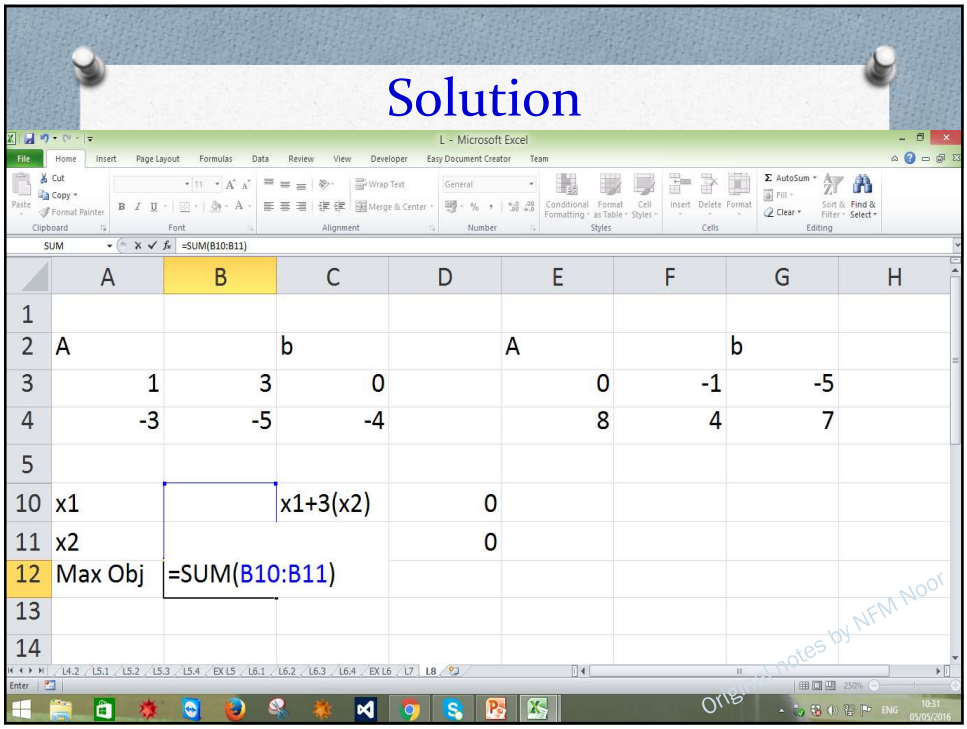
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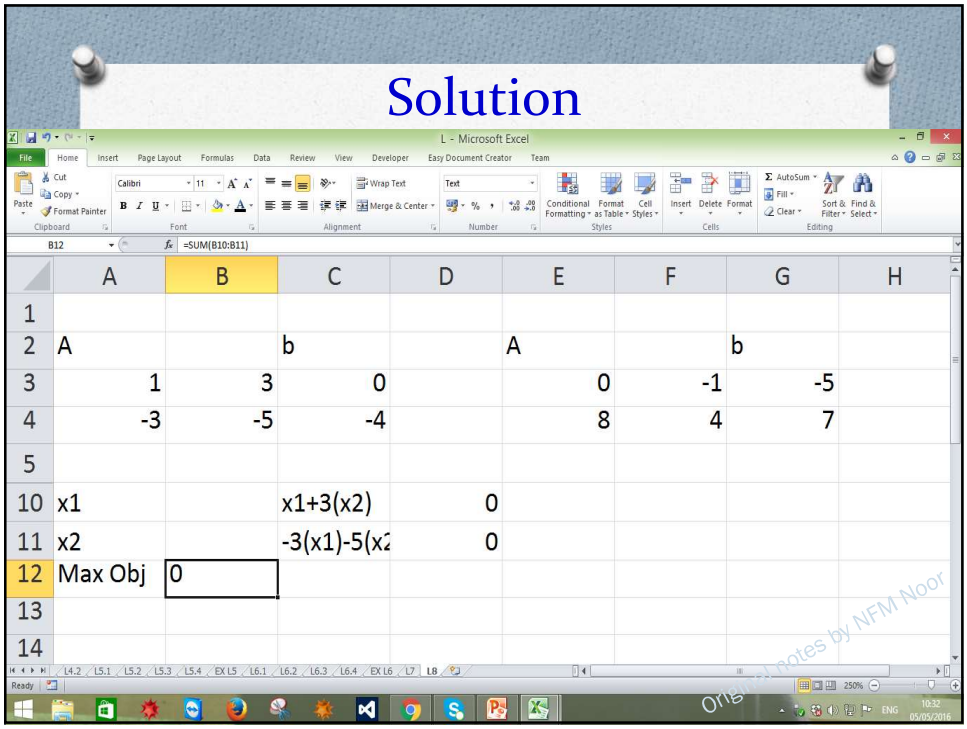
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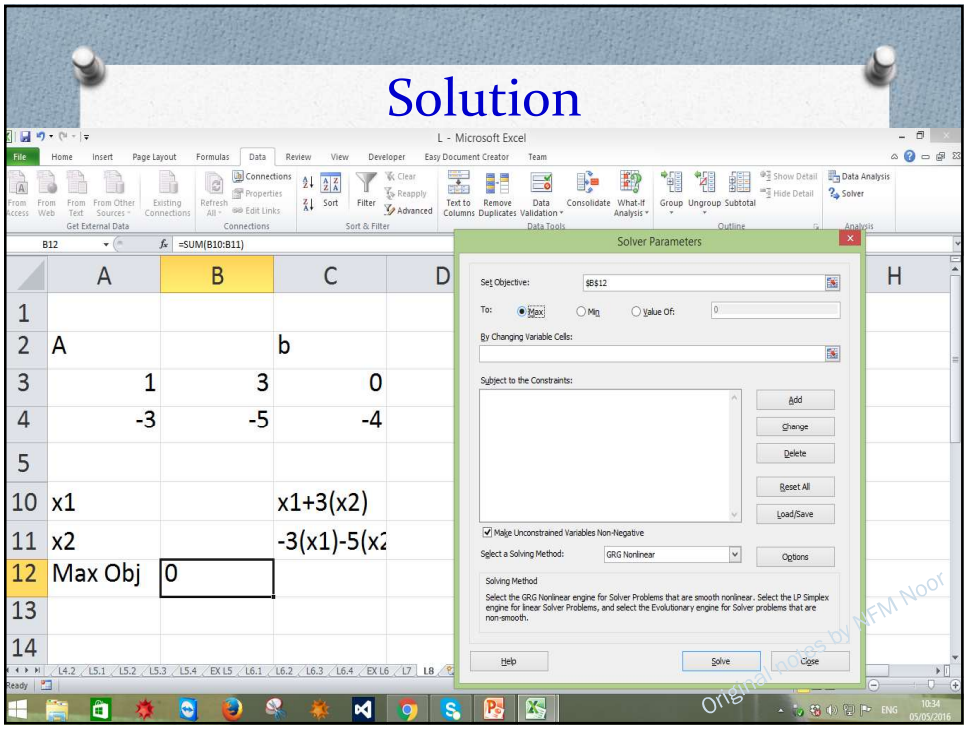
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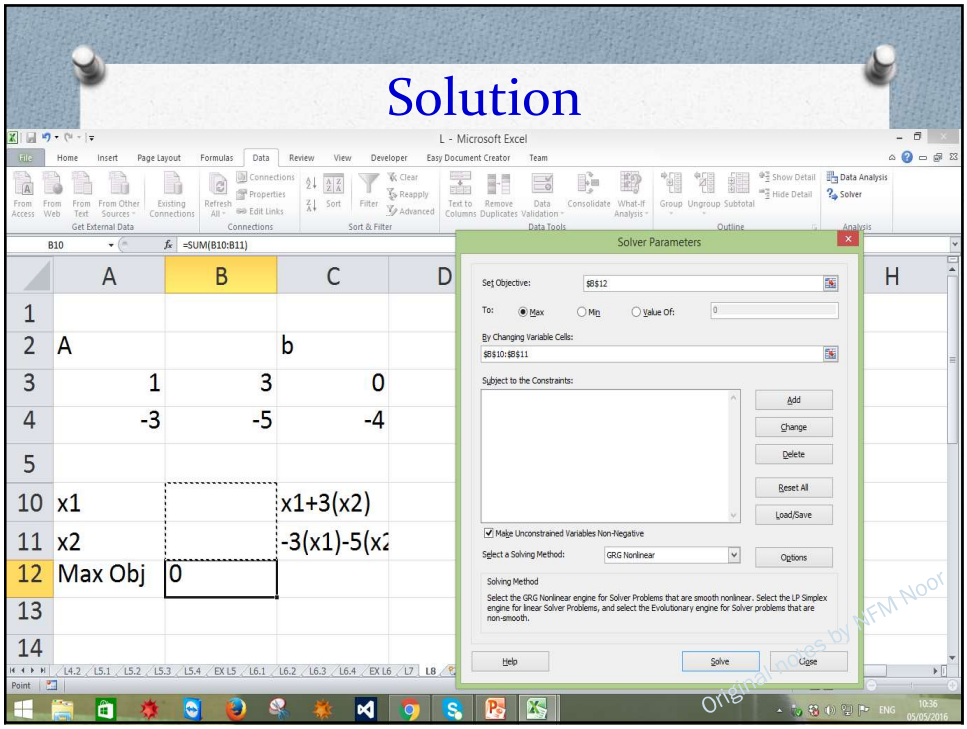
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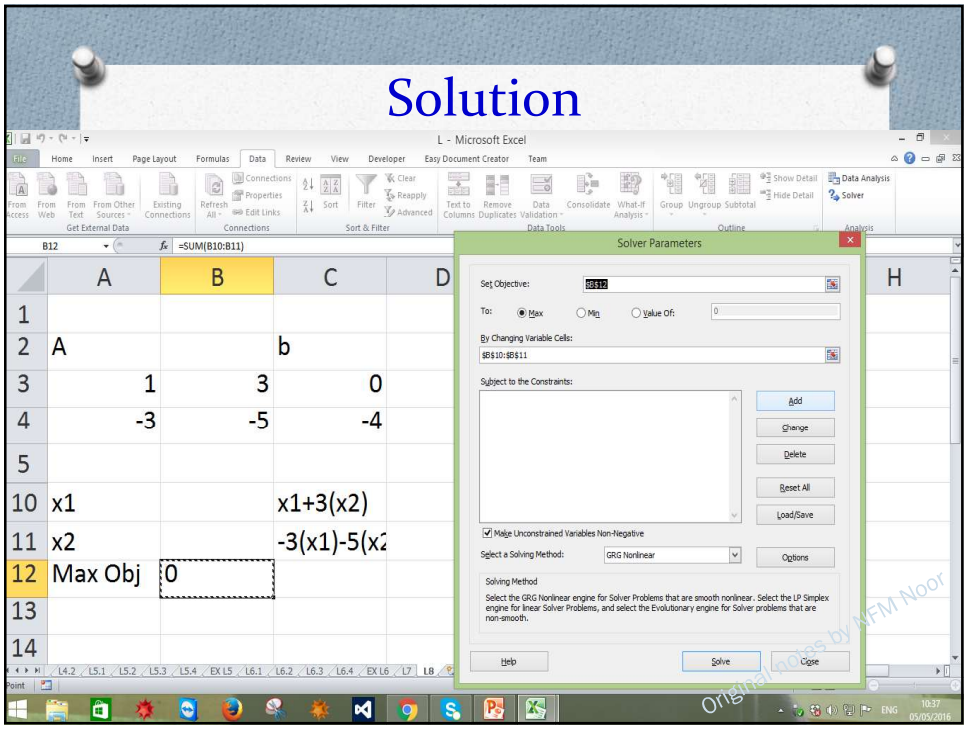
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Solution

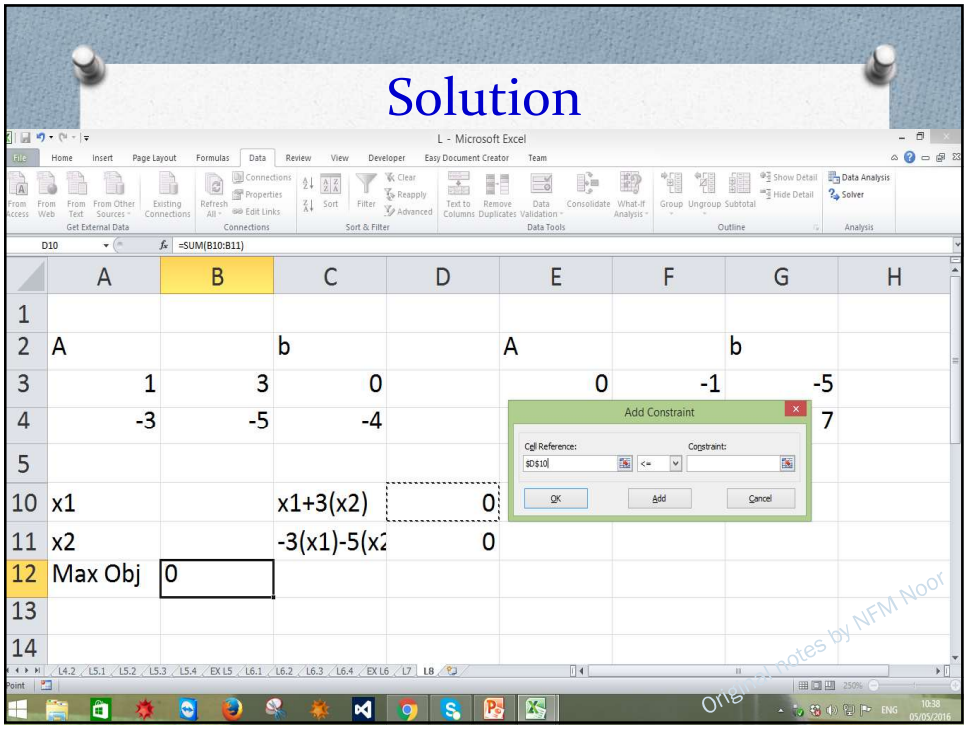


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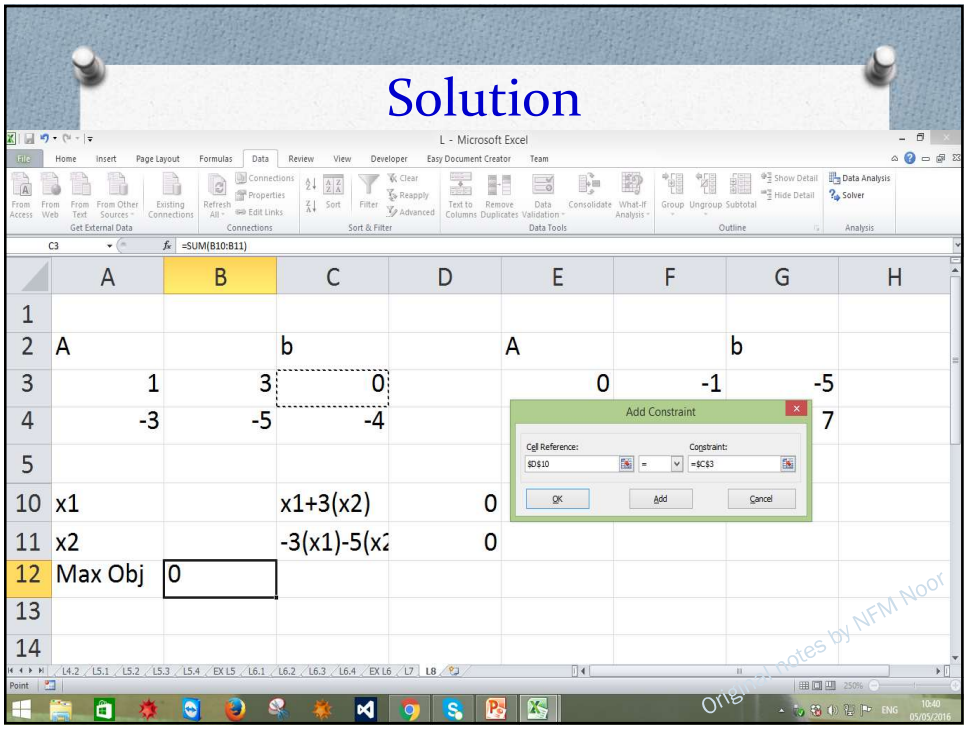
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2	A		b		
3	1	3	0		
4	-3	-5	-4		
5					
10	x1		x1+3(x2)		
11	x2		-3(x1)-5(x2		
12	Max Obj	0			
13					
14					

Solution



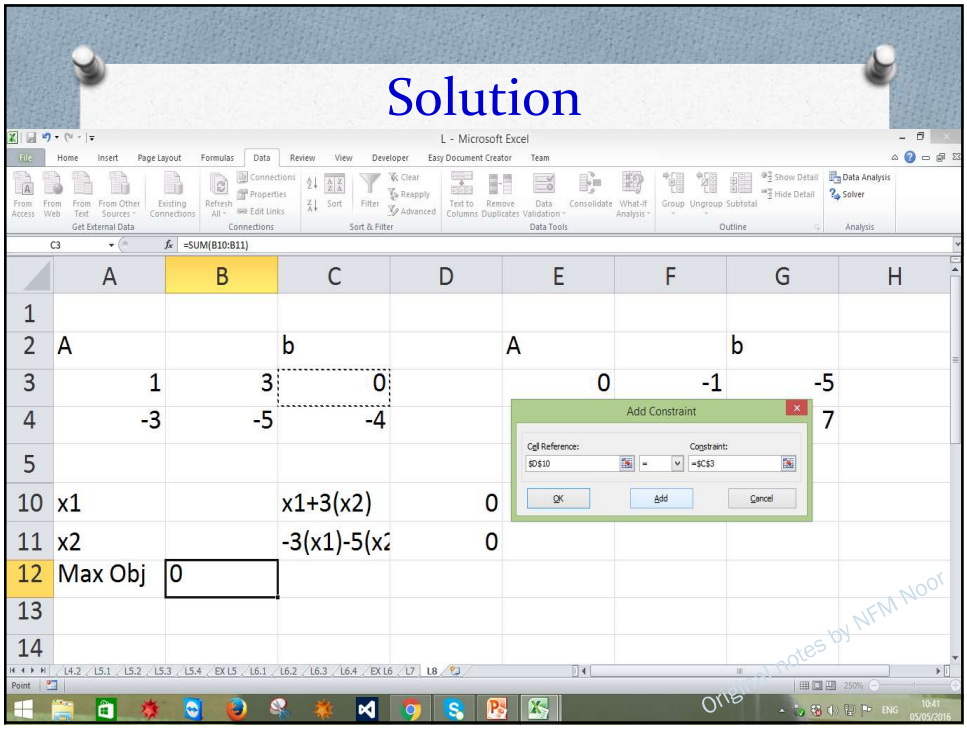
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2	A		b		A		b	
3	1	3	0		0	-1	-5	
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5								
10	x1		x1+3(x2)	0				
11	x2		-3(x1)-5(x2	0				
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Solution



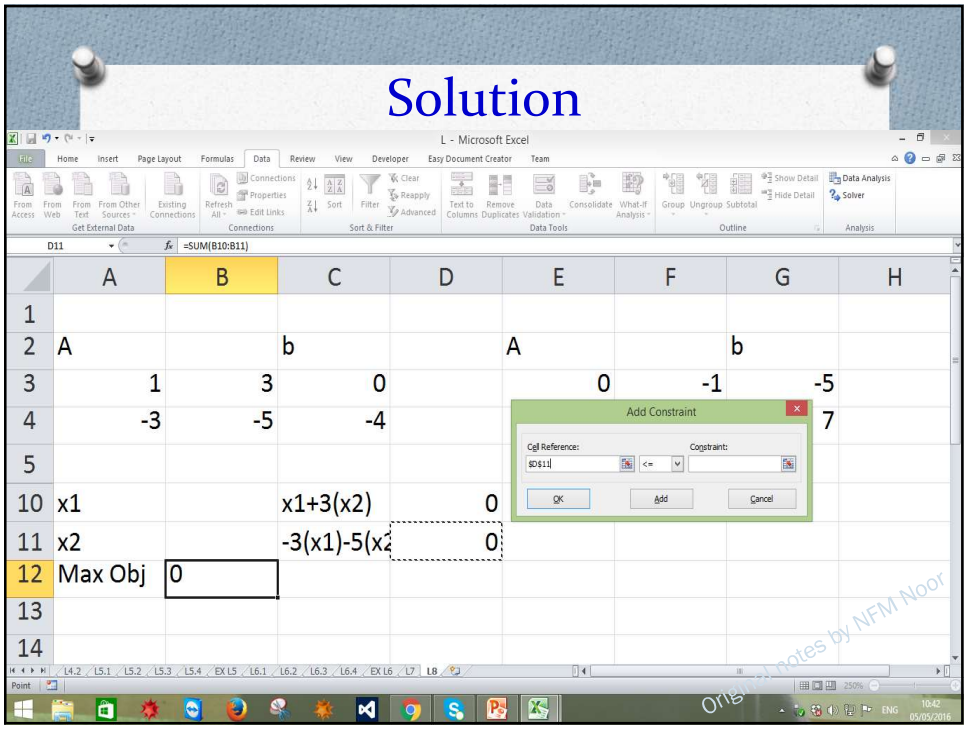
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2	A		b		A		b	
3	1	3	0		0	-1	-5	
4	-3	-5	-4				7	
5								
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11	x2		-3(x1)-5(x2	0				
12	Max Obj	0						
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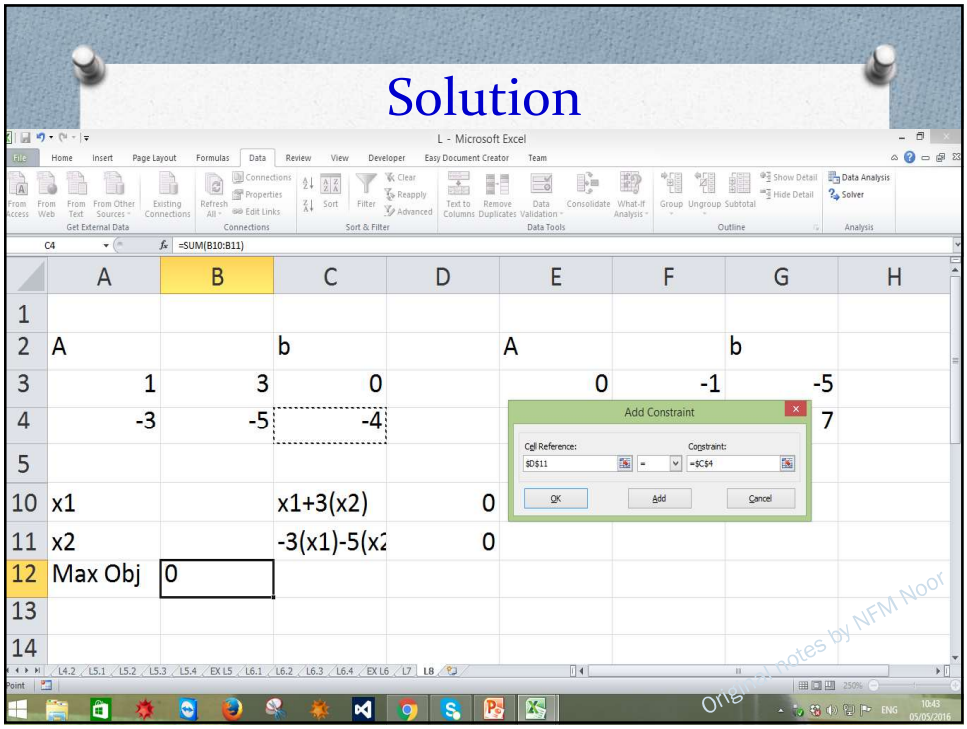
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2	A		b		A		b	
3	1	3	0		0	-1	-5	
4	-3	-5	-4				7	
5								
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11	x2		-3(x1)-5(x2	0				
12	Max Obj	0						
13								
14								

Solution



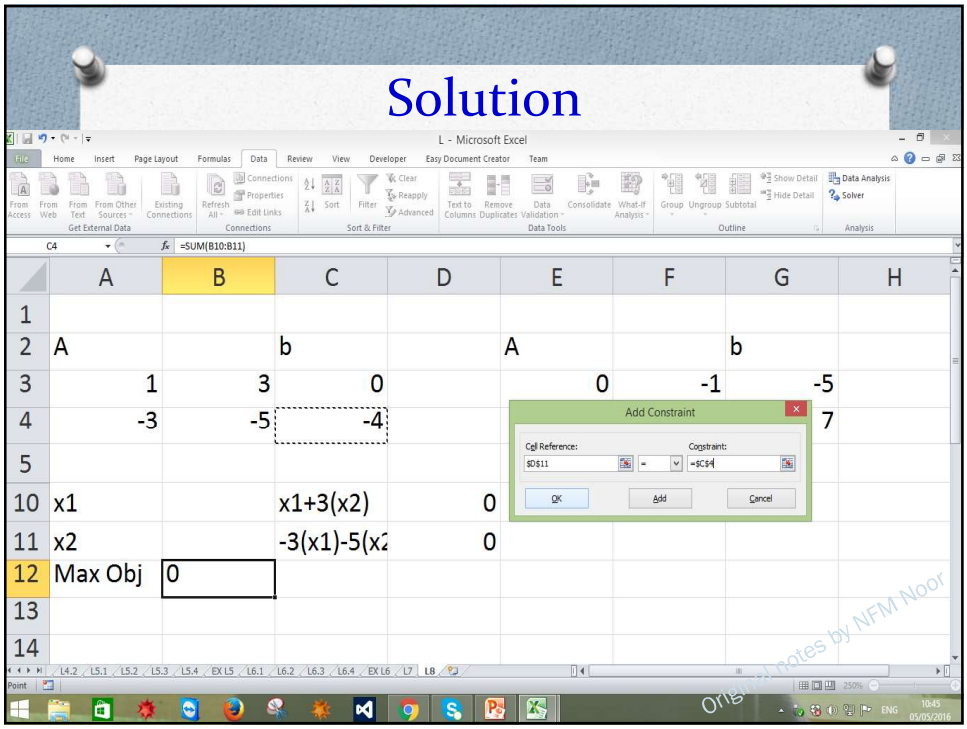
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2	A		b		A		b	
3	1	3	0		0	-1	-5	
4	-3	-5	-4				7	
5								
10	x1		x1+3(x2)	0				
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Solution

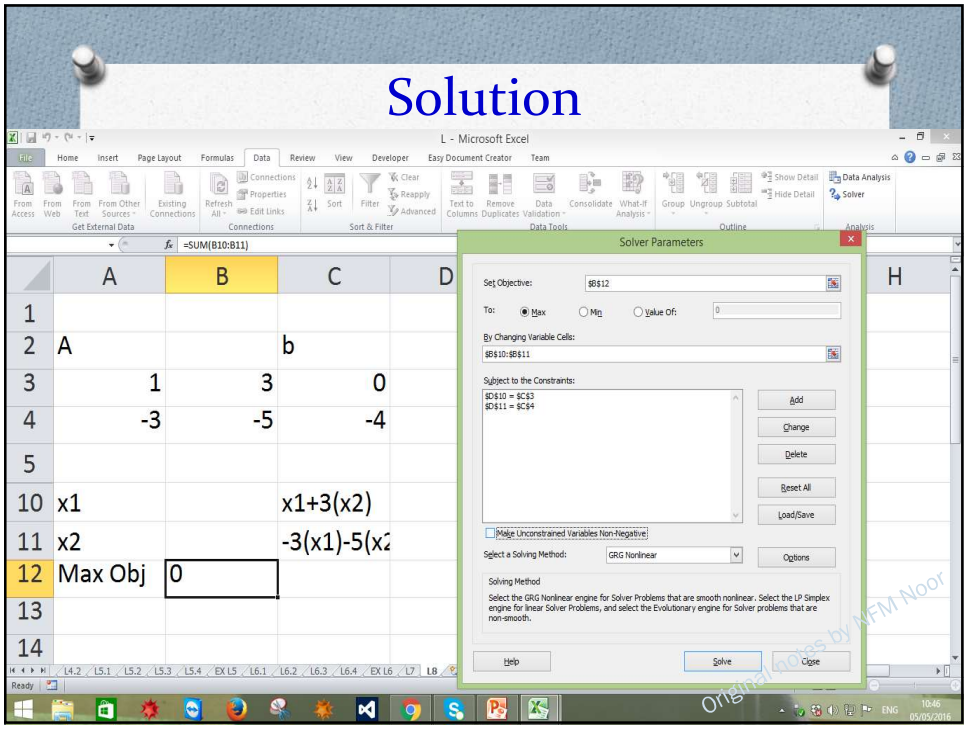


	A	B	C	D	E	F	G	H
1								
2	A		b		A		b	
3	1	3	0		0	-1	-5	
4	-3	-5	-4				7	
5								
10	x1		x1+3(x2)	0				
11	x2		-3(x1)-5(x2	0				
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14								

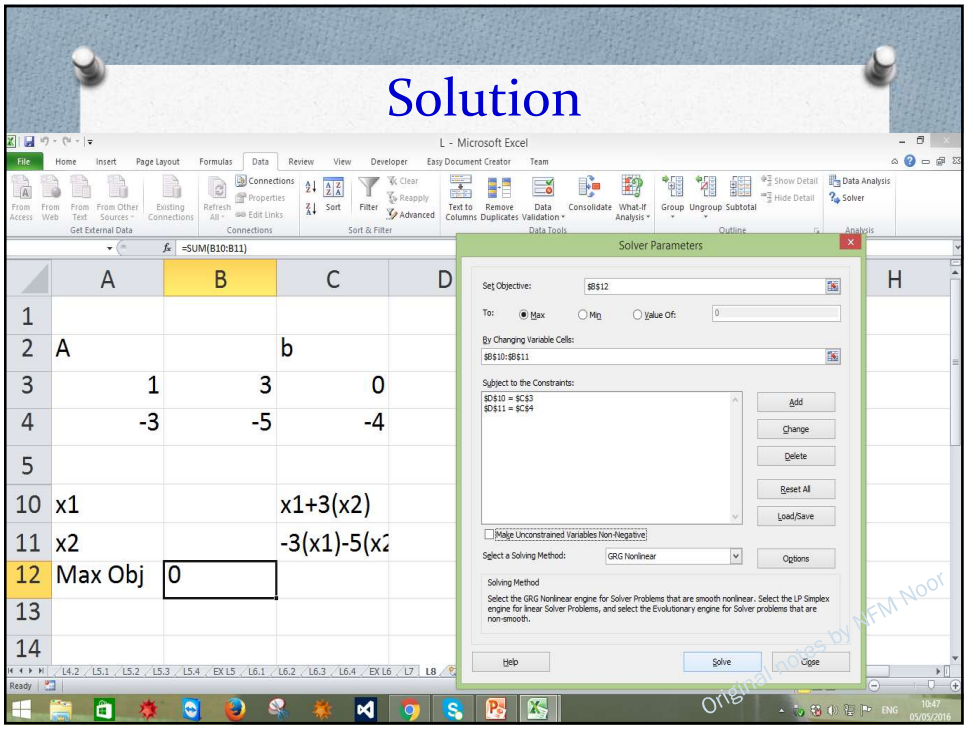
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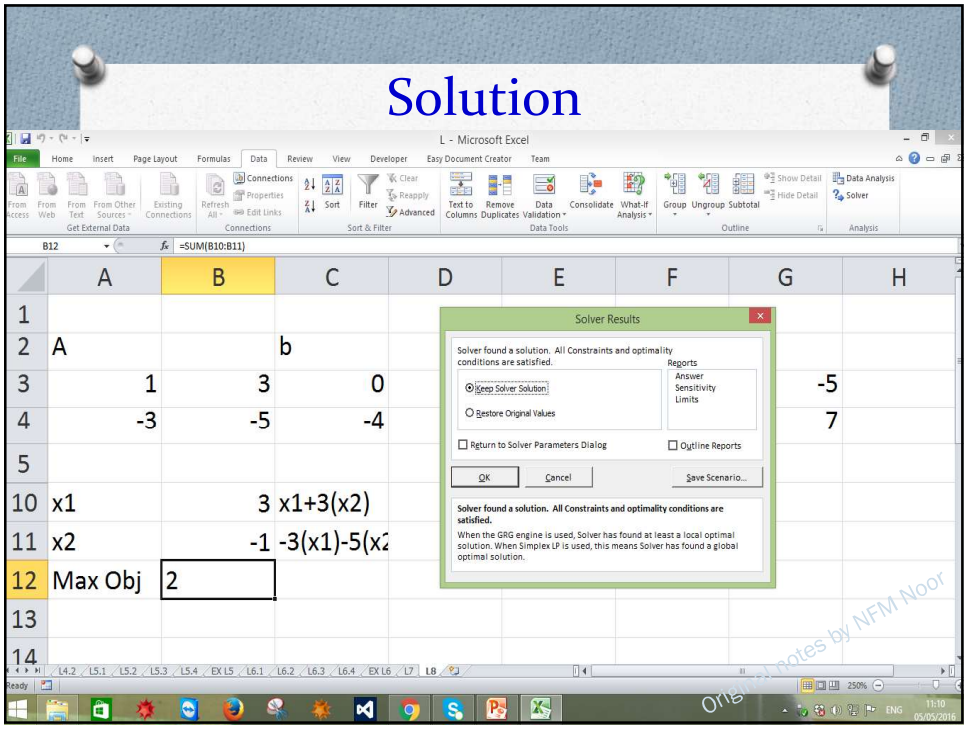
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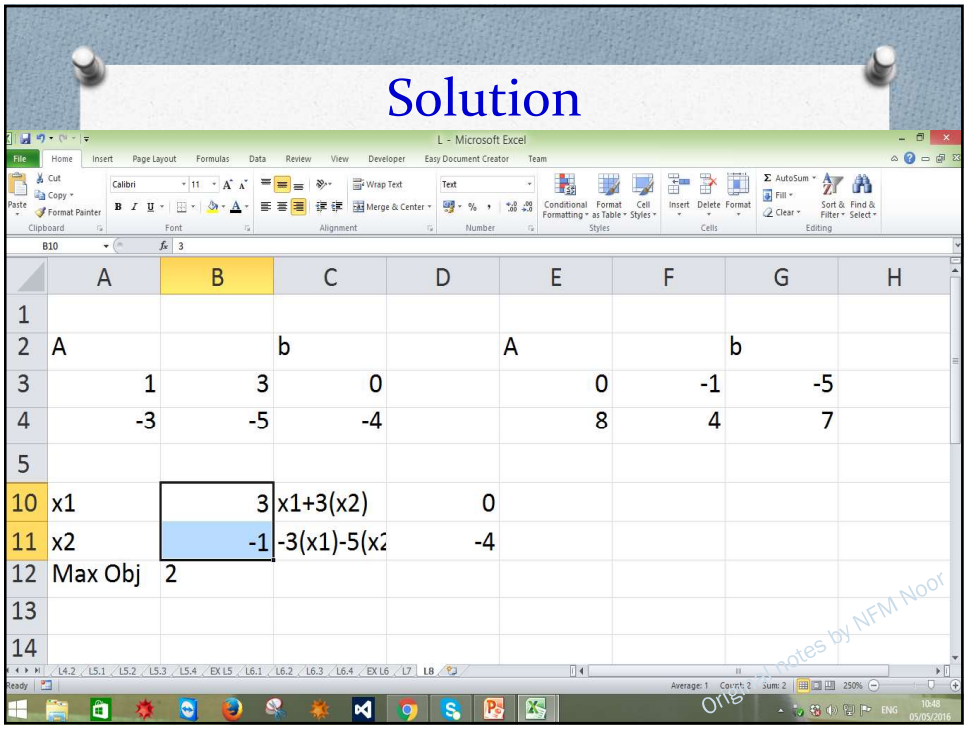
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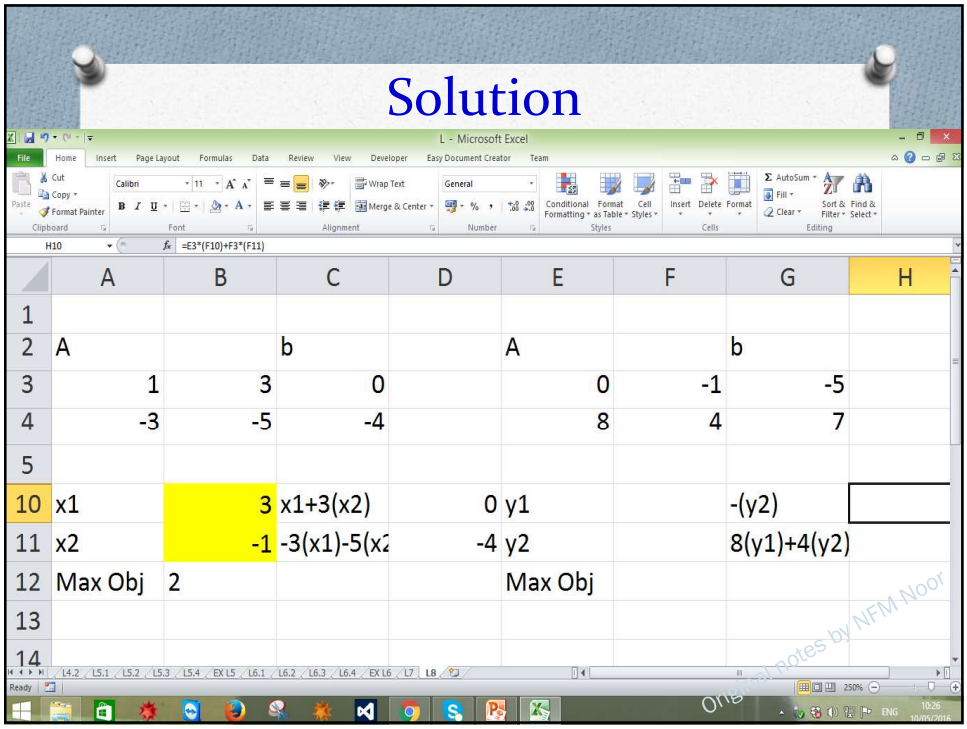
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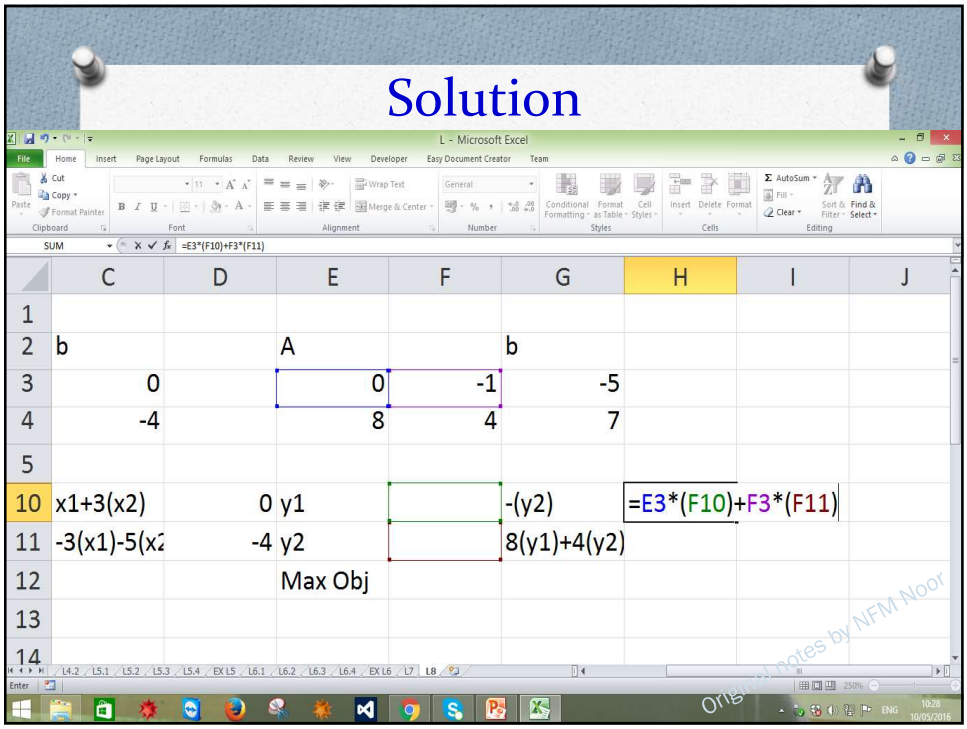
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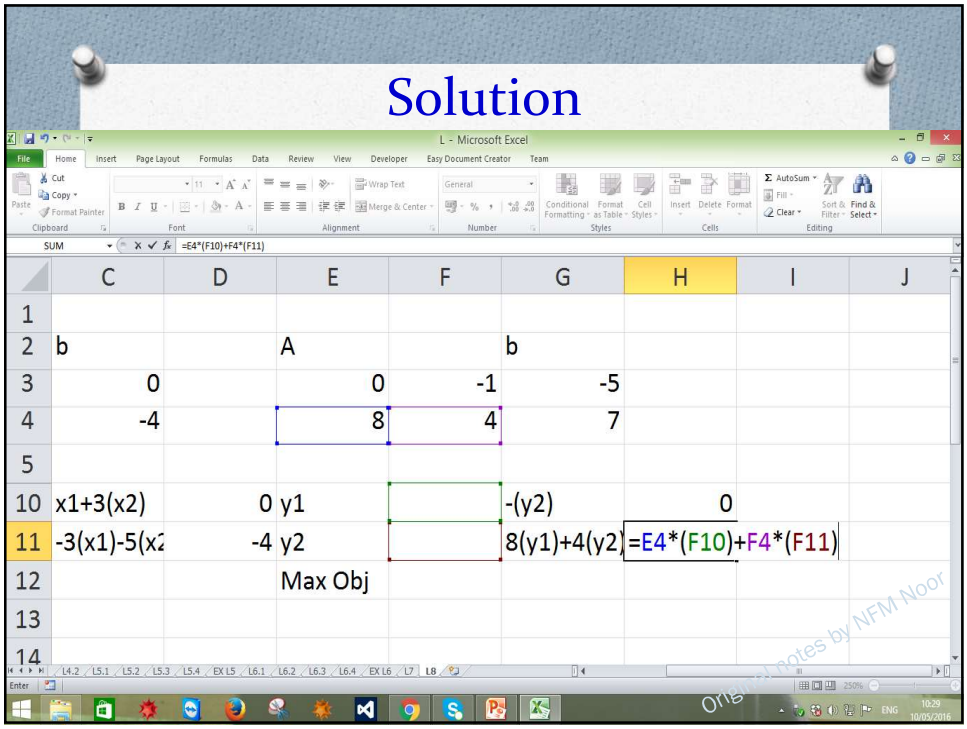
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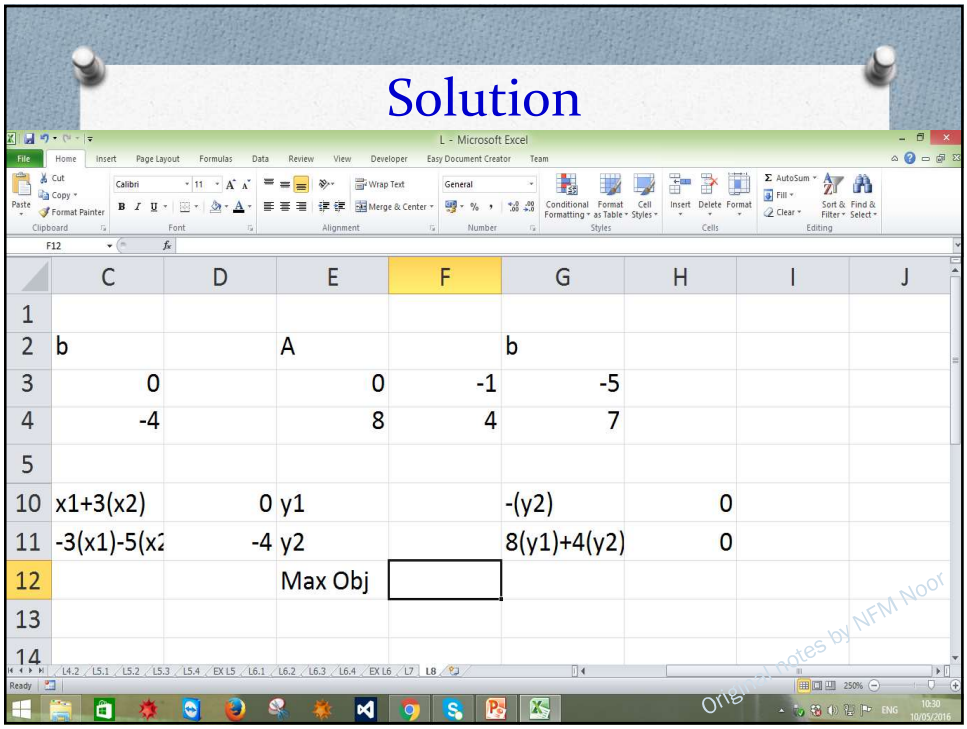
Solution



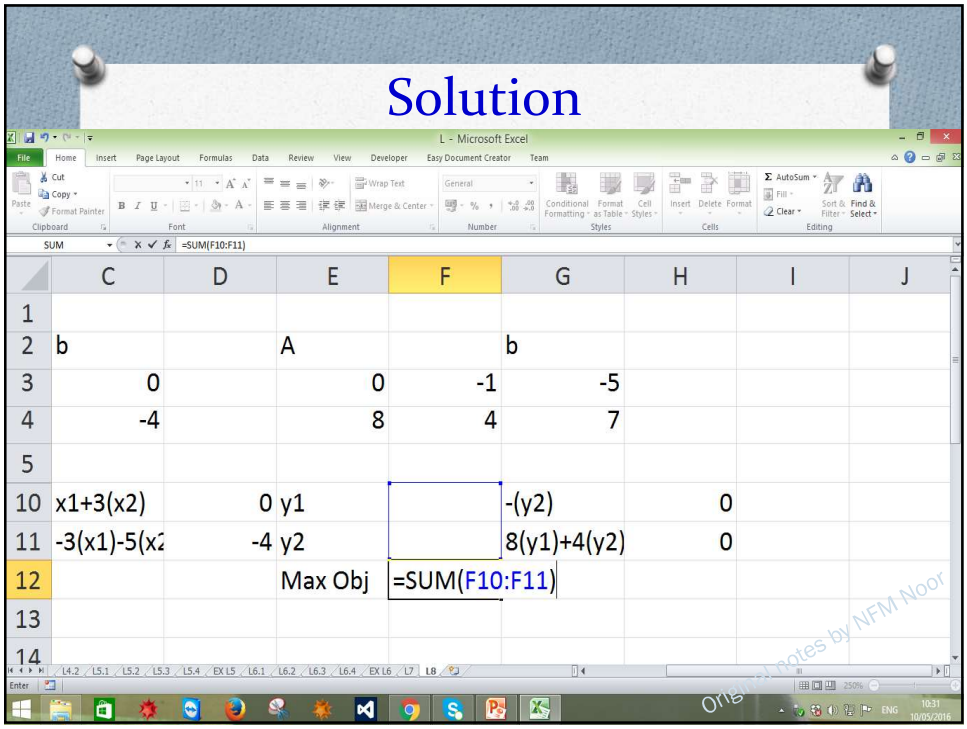
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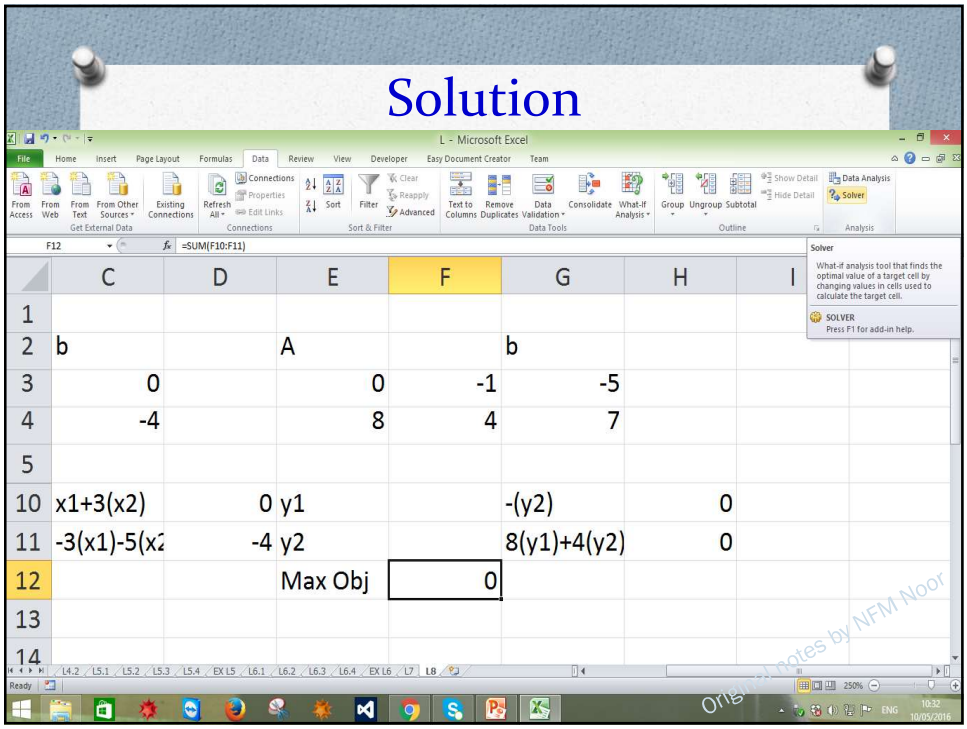
Solution



Solution

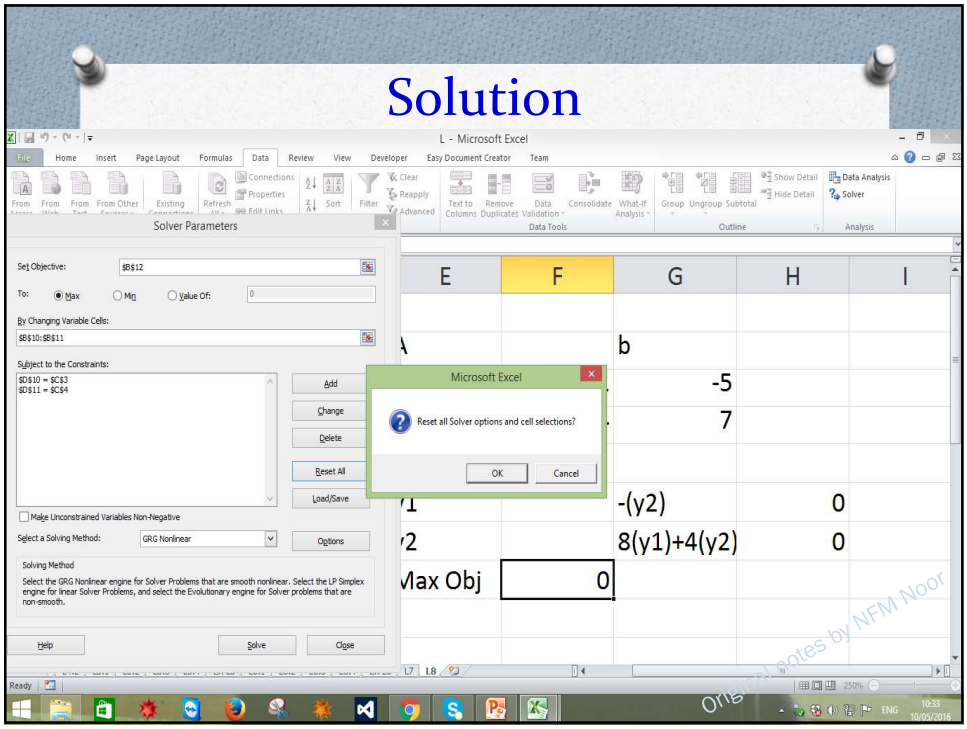


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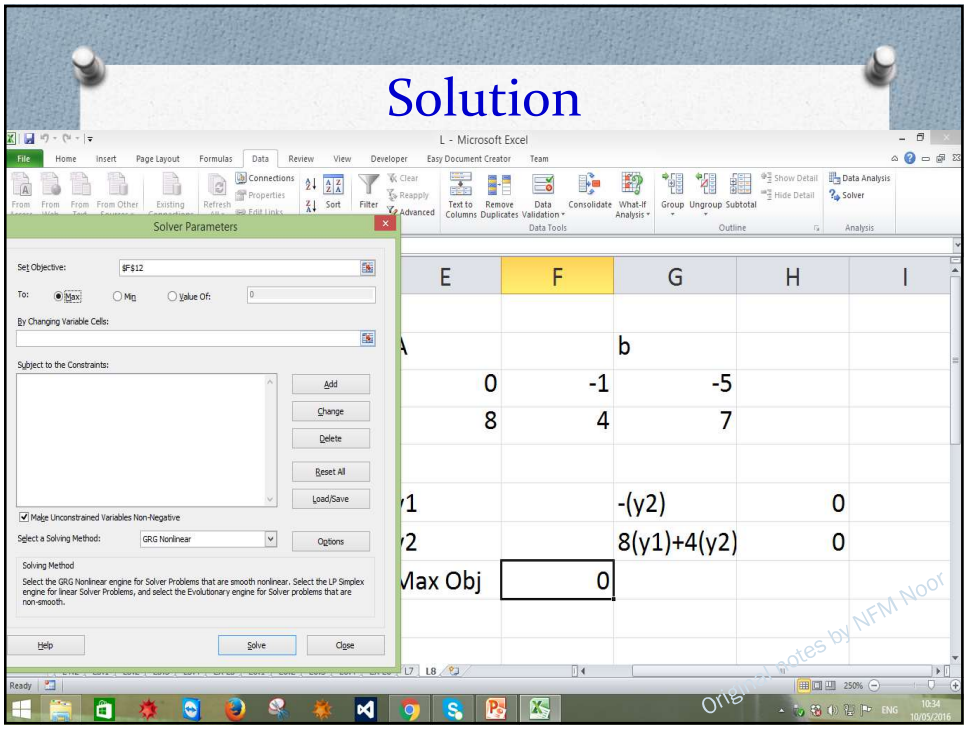
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1							
2	b		A		b		
3	0		0	-1	-5		
4	-4		8	4	7		
5							
10	x1+3(x2)	0	y1		-(y2)	0	
11	-3(x1)-5(x2	-4	y2		8(y1)+4(y2)	0	
12			Max Obj	0			
13							
14							

Solution

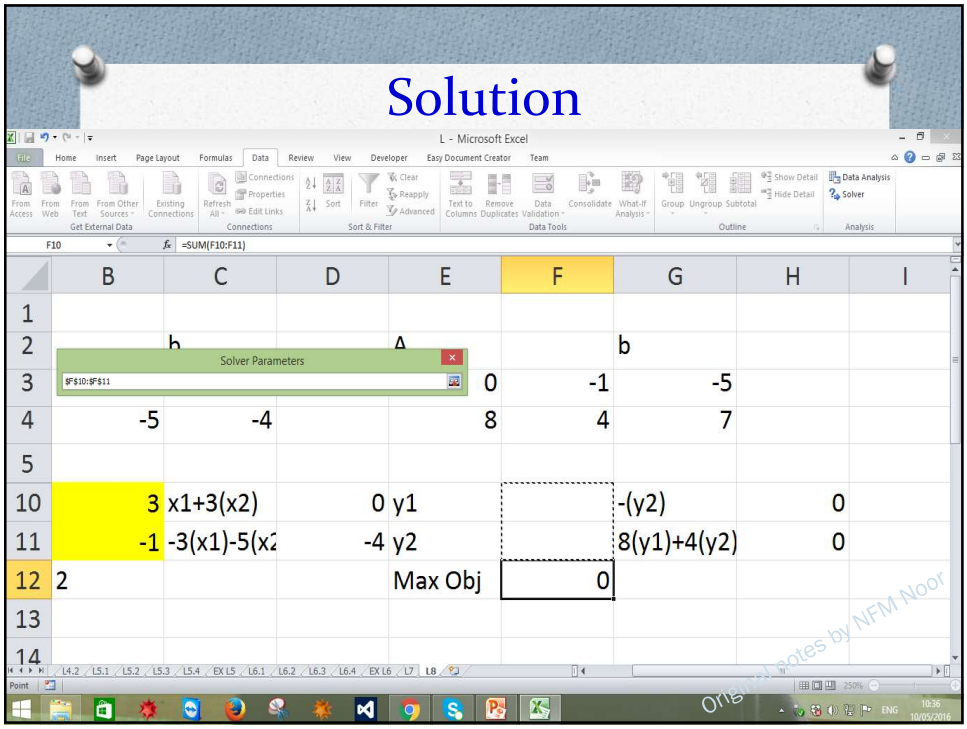


	E	F	G	H	I
			b		
			-5		
			7		
			-(y2)	0	
			8(y1)+4(y2)	0	
	Max Obj	0			

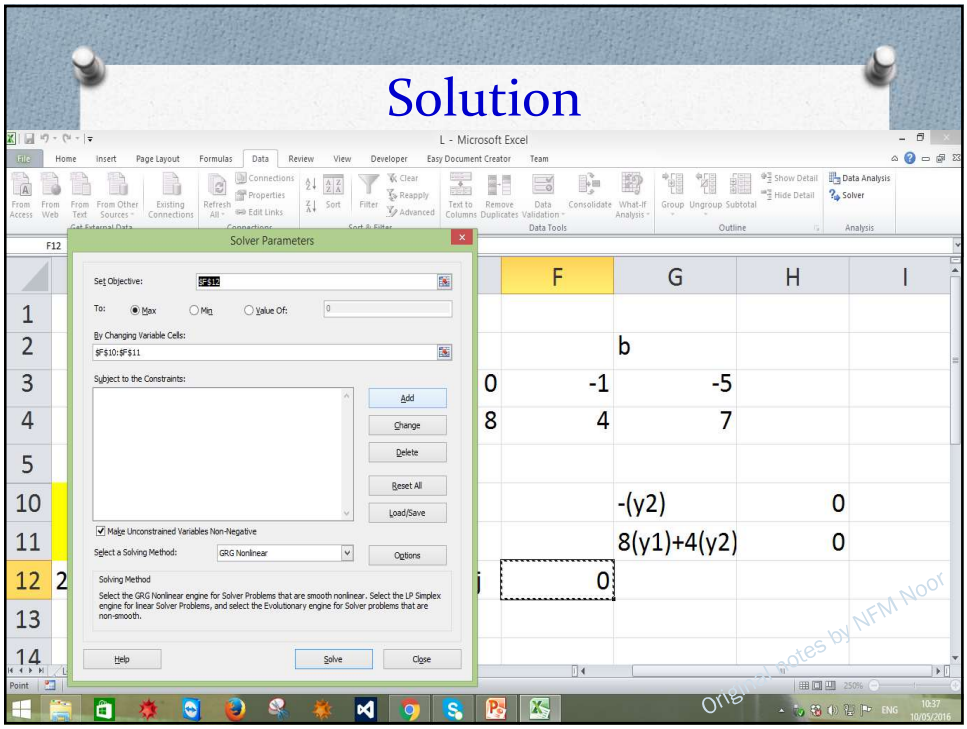
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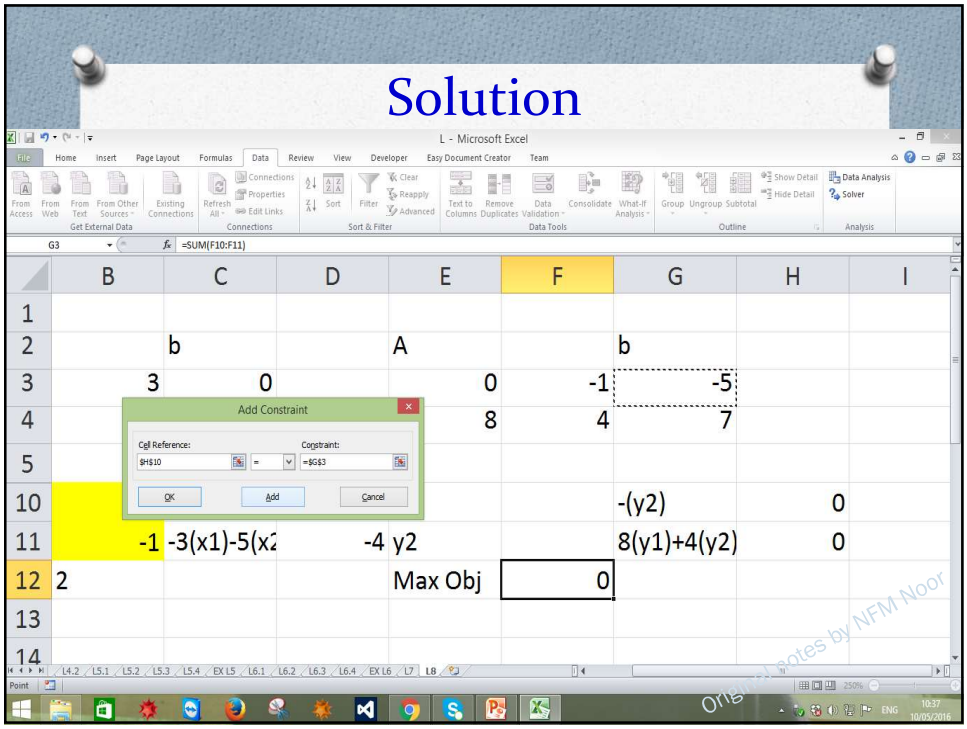
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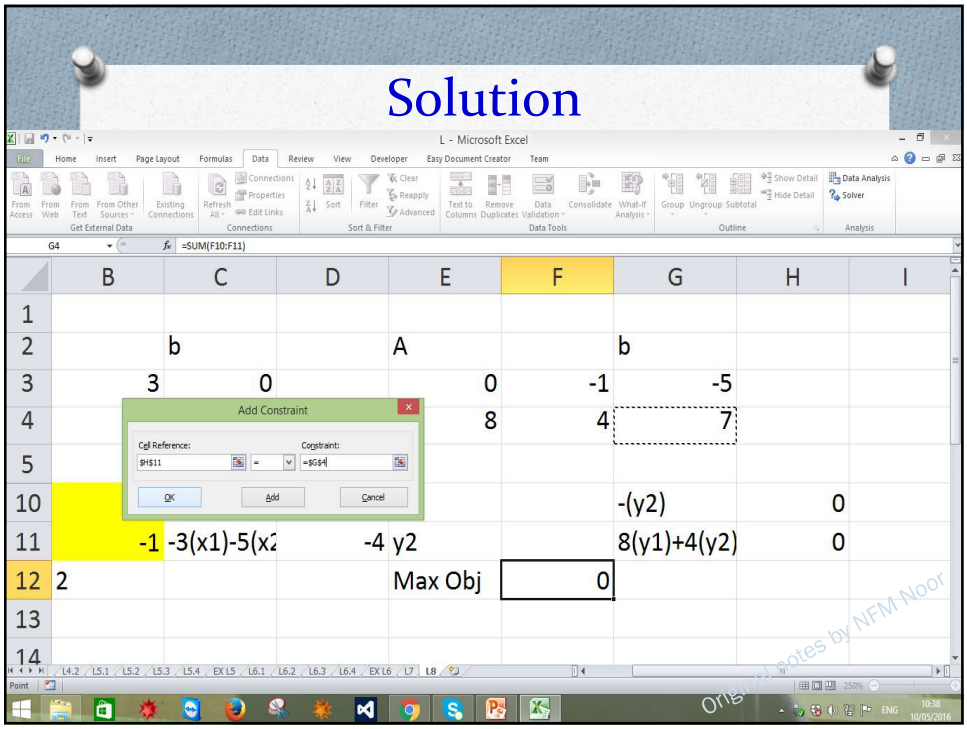
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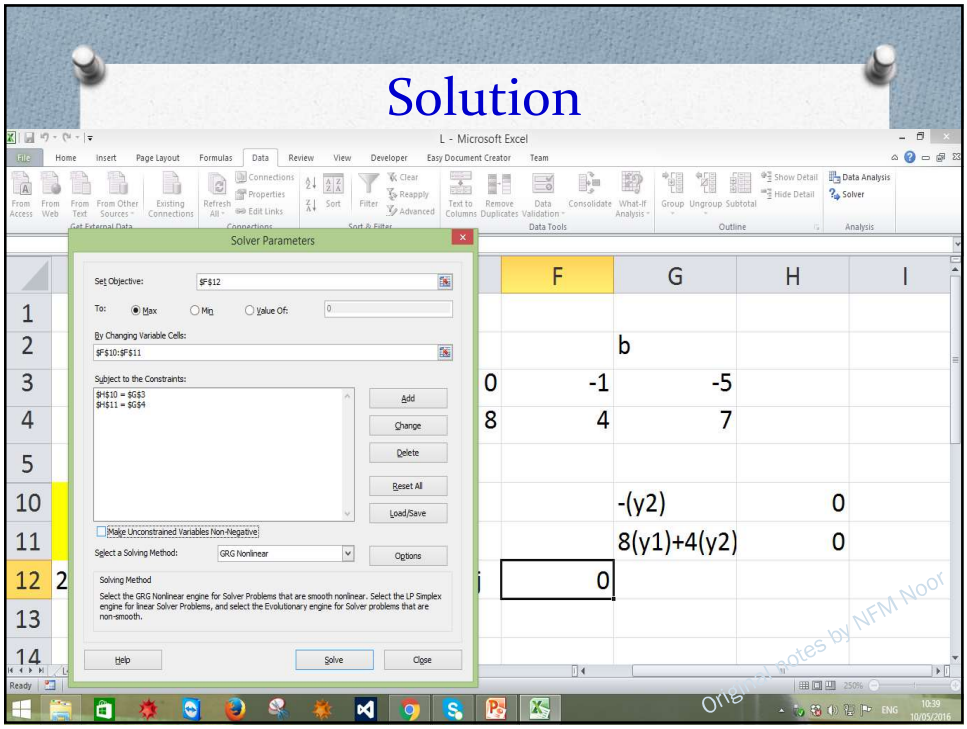
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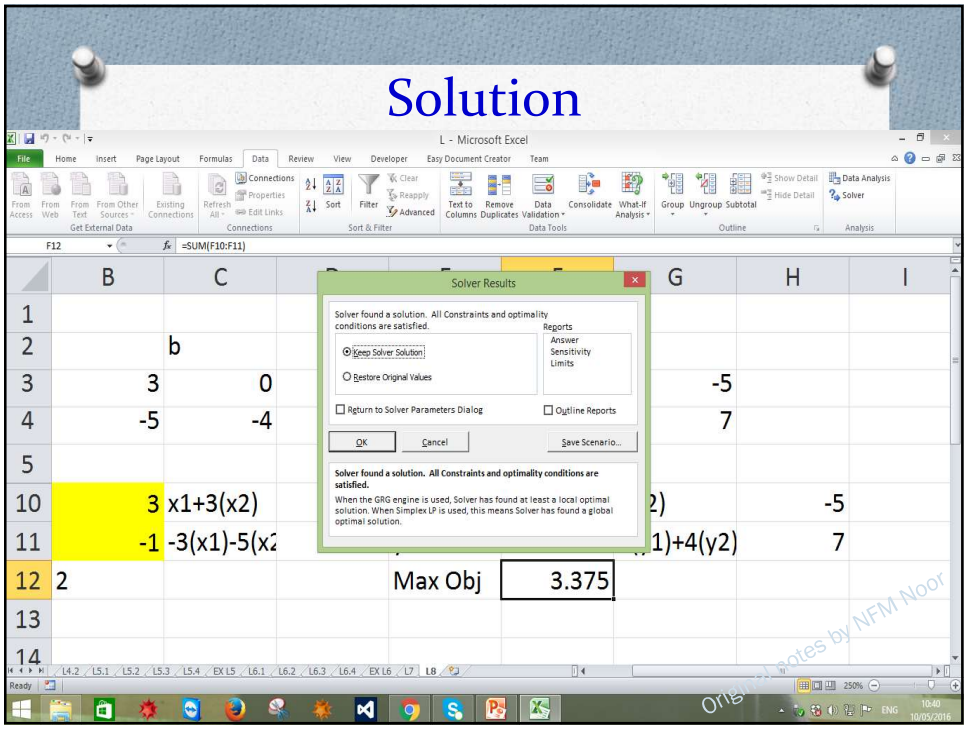
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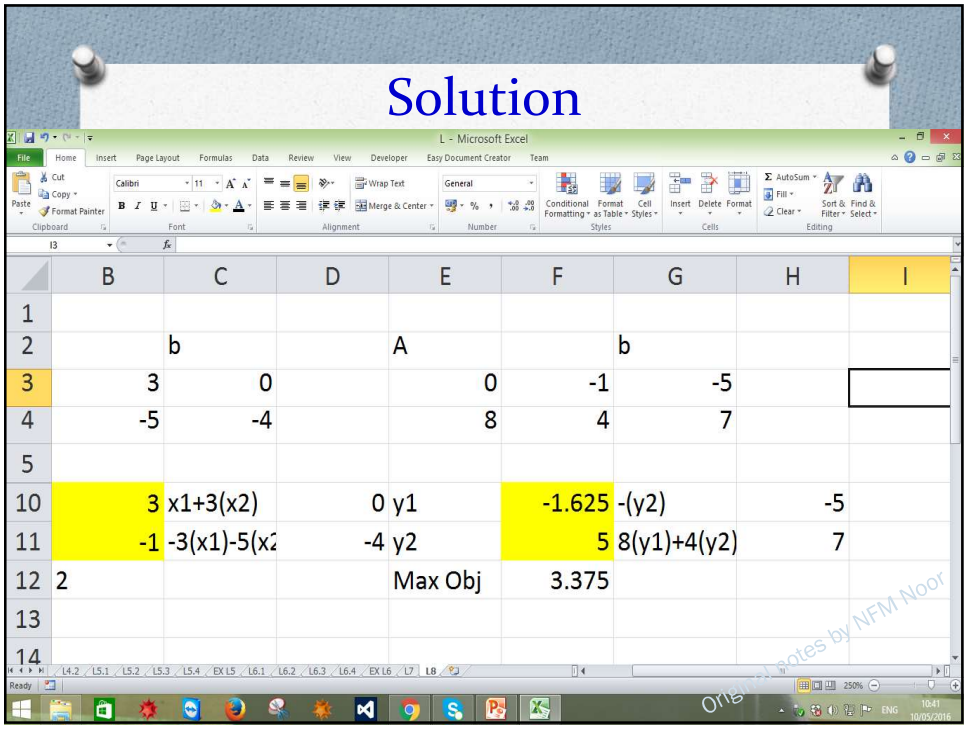
Solution



Solution



Solution



Exercise 2

Solve the following systems using Excel Solver:

1. $\begin{pmatrix} 4 & -5 & 0 \\ -1 & -2 & 9 \\ 3 & 3 & 8 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 12 \\ 3 \\ -9 \end{pmatrix}$
2. $\begin{pmatrix} -7 & 1 & 2 \\ 10 & 0 & -4 \\ 3 & -5 & -1 \end{pmatrix}\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 7 \end{pmatrix}$

Original notes by NFM Noor

Summary

- Excel provides tools for solving system of linear equations.
- Inverse method can be used to solve system of linear equations by using matrix operations.
- Excel solver can be used to solve system of linear equations by using simple linear programming approach.

Original notes by NFM Noor